

Aditya Kumar Singh

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Professional Summary

Linux Kernel and Embedded Systems Engineer with 5+ years of experience at Amazon and Qualcomm, specializing in Linux kernel, BSP, ARM platforms, embedded firmware, and wireless systems. Experienced in GPIO, pin control, I2C, TF-A, SCP firmware, ath11k/ath12k, IEEE 802.11, Wi-Fi 6/6E/7, and MLO, with strong expertise in C/C++, Python, and upstream Linux development.

Education

Indian Institute of Technology Madras

MASTER'S DEGREE IN COMPUTER SCIENCE AND ENGINEERING (SPL. IN INFORMATION SECURITY, WEB ENABLED)

Chennai, Tamil Nadu, India

Aug 2023–Jul 2026

- Secured a CGPA of 8.81 out of 10

National Institute of Technology Tiruchirappalli

BACHELOR'S DEGREE IN ELECTRICAL AND ELECTRONICS ENGINEERING

Tiruchirappalli, Tamil Nadu, India

Jul 2017–May 2021

- Secured a CGPA of 8.85 out of 10

Kendriya Vidyalaya Tatanagar (CBSE affiliated)

HIGHER SECONDARY EDUCATION

Jamshedpur, Jharkhand, India

Apr 2016–Mar 2017

- Secured 92.6%

Kendriya Vidyalaya Tatanagar (CBSE affiliated)

SECONDARY EDUCATION

Jamshedpur, Jharkhand, India

Apr 2015–Mar 2016

- Secured a CGPA of 10 out of 10 (95%) in Tenth grade

Achievements

- Received **Breakthrough Award** at Qualcomm (Apr 2025). The award recognizes employee's contributions to product innovation and cost saving initiatives.
- Received **Emerging Star Award** at Qualcomm (Oct 2023) and (Oct 2024). The award recognizes the employee who consistently go above and beyond their work and demonstrate reliability, perseverance, and focus on results.
- Received **Rockstar Rookie Award** at Qualcomm (Jan 2023). The award recognizes the best hires from college graduates for their exemplary performance and who has significantly contributed to Qualcomm's success in very short span of time and embraced the Qualcomm values.
- Secured **1st place** in Elecart and SCDC events at **IIT-M Shaastra'20**, a National Level Techno-management fest.
- Secured **4th rank** in **eYRC-2018**, a National Level Robotics Competition organized by **e-Yantra, IIT Bombay**.
- Finalist** of **Sangam**, an intra-college **Hardware Hackathon in Pragyan'19** (The International Techno-Management Festival of NIT-Trichy).

Work Experience

Software Development Engineer (SDE) II

Amazon Development Center India Pvt. Ltd

Kernel Developer

Dec 2025 – Present

- Core member of the BSP team developing Linux kernel support for Amazon's first-party ARM-based silicon platforms powering Echo, Fire TV, and Kindle devices.
- Own and develop Linux kernel subsystems including Pin Control, GPIO, and I2C, enabling board bring-up and peripheral integration.
- Work across the ARM boot and firmware stack, including BootROM development, Trusted Firmware-A (TF-A), and SCP firmware, with focus on boot flow and early platform initialization.

Senior Software Engineer

Qualcomm India Pvt. Ltd.

Promoted from Associate Software Engineer -> Software Engineer -> Senior Software Engineer

Linux Wireless Driver Developer

Jun 2021 – Nov 2025

- Worked on IEEE 802.11 standards including 11be (Wi-Fi 7) and 11ax (Wi-Fi 6), contributing to ath11k and ath12k Linux wireless driver development.
- Contributed to userspace wireless components including hostapd, wpa_supplicant, and iw, alongside kernel driver development.
- Worked across key components of the wireless stack, including regulatory, MLME, and data path.
- Enabled Multi-Link Operation (MLO) in the upstream ath12k driver.
- Implemented Wi-Fi 6E support in the ath11k driver.
- Collaborated with customers including Google and Eero (Amazon) to support product launches based on ath11k/ath12k platforms.
- Open-Source Contributions:**
 - Linux Wireless Patchwork: [Link 1](#) | [Link 2](#)
 - Git repo: [ath](#) | [hostapd](#) | [iw](#)

Summer Internship at Hewlett Packard Enterprise

HPE

Vega

May 2020 – Jul 2020

- Evaluated Kafka, Redis, and Elasticsearch for an AI-driven primary storage management system by benchmarking their performance across different workloads and configurations.
- Developed Go-based interfaces using client libraries for Kafka, Redis, and Elasticsearch to automate data collection and performance evaluation.

Summer Internship under Prof. Kavi Arya at e-Yantra, IIT Bombay

e-Yantra

Bio-Inspired Sub—Carangiform FishBOT

May 2019 – Jul 2019

- Designed and developed a bionic sub-carangiform fish robot capable of forward motion and turning with variable radius and speed, controlled remotely.
- Built a cost-effective robotic platform with approximately 30 minutes of battery backup and demonstrated basic fish-like swimming capabilities.
- **GitHub Link:** <https://github.com/adityasingh3007/Bio-Inspired-Sub-Carangiform-FishBOT>

Projects

Automation system for easy upstreaming of patches

Qualcomm

Python, Shell

Jul 2024 – Nov 2025

- Architected and maintained a build automation framework inspired by OpenWrt to generate customized Linux-based images from upstream repositories.
- Automated integration of private patches, kernel and firmware customization, and image deployment and validation on embedded target devices.
- Improved build reproducibility and reduced manual effort, accelerating the Linux kernel upstreaming and validation workflow.

E-mail bot to crawl internal mailing list

Qualcomm

Python, Shell

Jan 2025 – Nov 2025

- Designed and developed an internal automation bot for wireless kernel patch validation and review, integrating mailing lists and patch submission platforms with automated kernel testing workflows.
- Automated patch application to appropriate kernel trees and validation using checkpatch, cross-compilation, 32-bit builds, allmodconfig, and static analysis.
- Automated reporting of validation results by posting test outcomes directly to the corresponding patch emails, reducing manual effort in the patch review cycle.

AntBOT

e-YRC - 2018

Image Processing, 3D Designing, PID Control, Path Planning Algorithms

Nov 2018 – Mar 2019

- Developed AntBOT, a robotics platform inspired by ant foraging behavior, capable of collecting and transporting objects while following a predefined path.
- Implemented path-following and task execution to enable the robot to navigate between designated areas and perform collection and disposal tasks.
- **GitHub Link:** <https://github.com/adityasingh3007/AntBOT>

Skills

- **Programming:** C, C++, Embedded C, Python, Shell
- **Linux & Embedded:** Linux Kernel, Embedded Linux, BSP, ARM, OpenWrt, Buildroot, Yocto
- **Wireless & Networking:** IEEE 802.11, Wi-Fi 6/6E/7, ath11k, ath12k, MLO, hostapd, wpa_supplicant, iw
- **Firmware:** Trusted Firmware-A (TF-A), SCP
- **Development Tools:** Git, Gerrit
- **Languages:** English, Hindi

Position of Responsibility

President, Spider R&D Club of NIT-Trichy:

Apr 2020 – May 2021

Member since Jul 2018

Led technical activities and projects in Embedded Systems, Robotics, Control, Signal Processing, and Machine Learning through national and college-level competitions.

Workshop Coordinator, Currents:

Aug 2018 – Apr 2021

Organized and conducted technical workshops as part of Currents, the annual Electrical and Electronics Engineering symposium.

Extracurricular Activity

- Won 1st Prize in Carrom in Sportsfete-18, an inter-department sports fest of college.
- Part of the NSO team of NIT Trichy.